Exercise 1.4

1	2	3	4	5	6	7	8	9	10	11	12	13
(d)	(b)	(a)	(c)	(d)	(b)	(b)	(b)	(a)	(c)	(c)	(b)	(c)
14	15	16	17	18	19	20	21	22	23	24	25	
(b)	(c)	(c)	(b)	(b)	(a)	(a)	(b)	(c)	(d)	(c)	(a)	

Miscellaneous problems

1. $\rho(A)=2$ 2. $\rho(A)=3$ 3. $\rho(A)=3$
4. The given system is inconsistent and has no solution. 5. $k = 8$.
6. The equations are inconsistent when k assumes any real value other than 0.
7. $x=1,\ y=2$ and $z=2$
8. Cost of wheat is ₹30/kg, Cost of sugar is ₹40/kg and Cost of rice is ₹50/kg.
9. The rates of commission for A,B and C are ₹2, ₹4 and ₹11 respectively
10. 39%

2. Integral Calculus – I

Exercise: 2.1

1. $\frac{2}{9}(3x+5)^{\frac{3}{2}}+c$ 2. $\frac{81x^5}{5}-\frac{16}{3x^3}-72x+c$ 3. $6x-\frac{13x^2}{2}-\frac{5x^3}{3}+c$
4. $\frac{2x^{\frac{9}{2}}}{9}-\frac{4x^{\frac{5}{2}}}{5}+2x^{\frac{3}{2}}+c$ 5. $\frac{(4x+7)^{\frac{3}{2}}}{3}-\frac{(4x+7)^{\frac{1}{2}}}{2}+c$
6. $\frac{1}{3}\left[(x+1)^{\frac{3}{2}}-(x-1)^{\frac{3}{2}}\right]+c$ 7. $b=\frac{13}{2}, c=-2, f(x)=\frac{x^2}{2}+\frac{13}{2}x-2$
8. $c=-20, f(x)=2x^4-x^2-20$

Exercise: 2.2

1. $x^2+\frac{1}{2}\log|x|-2x+c$ 2. $\frac{x^4}{4}+\frac{x^3}{3}+2\log|x-1|+c$
3. $\frac{x^3}{3}-x^2+4x-8\log|x+2|+c$ 4. $\frac{x^3}{3}-x^2+3x-4\log|x+5|+c$
5. $11\log|x-3|-8\log|x-2|+c$ 6. $\log|x+1|+3\log|x-3|+\frac{2}{(x+1)}+c$
7. $\log|x^3-x^2+5x-5|+c$ 8. $c=\frac{\pi}{4}, f(x)=\log|x|+\frac{\pi}{4}$

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